

Additive regression for non-Euclidean responses and predictors

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1 The problem

Many nonparametric regression methods are designed for Euclidean responses and predictors. This paper studies additive regression in a general setting where the response takes values in a separable Hilbert space and the predictors take values in semimetric spaces. The framework includes several types of non-Euclidean data, such as functional, density-valued, compositional, spherical, toroidal, and shape data.

2 The setup

Let $Y \in \mathbb{H}$ be a Hilbert-space-valued response and let

$$X = (X_1, \dots, X_d) \in \mathcal{X} := \prod_{j=1}^d \mathcal{X}_j,$$

where X_j takes values in the semimetric space $(\mathcal{X}_j, d_j)$.

The additive model is

$$Y = E(Y) \oplus \sum_{j=1}^d f_j(X_j) \oplus \epsilon,$$

where $f_j : \mathcal{X}_j \rightarrow \mathbb{H}$ are unknown component maps and

$$E(\epsilon \mid X) = 0.$$

For identifiability, the components satisfy

$$E\{f_j(X_j)\} = 0, \quad 1 \leq j \leq d.$$

The goal is to estimate the full additive regression map

$$f(x) = E(Y) \oplus \sum_{j=1}^d f_j(x_j)$$

and the individual component maps f_j .

3 The main idea

The paper extends smooth backfitting to Hilbert-space-valued responses and semimetric-space-valued predictors. Let

$$m_j(x_j) = E(Y \mid X_j = x_j)$$

be the marginal regression map. Taking conditional expectation of the additive model given $X_j = x_j$ yields

$$f_j(x_j) = m_j(x_j) \ominus E(Y) \ominus \sum_{k \neq j} \int_{\mathcal{X}_k} f_k(x_k) \frac{p_{jk}(x_j, x_k)}{p_j(x_j)} d\mu_k(x_k), \quad 1 \leq j \leq d,$$

where p_j and p_{jk} are the marginal and joint densities of X_j and (X_j, X_k) , respectively. Thus, the component maps are characterized by a system of Bochner integral equations.

The unknown quantities are replaced by kernel-type estimators on the predictor spaces. The estimated system is

$$\hat{f}_j(x_j) = \hat{m}_j(x_j) \ominus \bar{Y} \ominus \sum_{k \neq j} \int_{\mathcal{X}_k} \hat{f}_k(x_k) \frac{\hat{p}_{jk}(x_j, x_k)}{\hat{p}_j(x_j)} d\mu_k(x_k), \quad 1 \leq j \leq d.$$

The solution

$$(\hat{f}_1, \dots, \hat{f}_d)$$

is the semimetric Bochner smooth backfitting estimator.

Because the system has no closed-form solution, the paper uses an iterative backfitting algorithm. Starting from initial estimators $\hat{f}_j^{[0]}$, the r -th update is

$$\begin{aligned} \hat{f}_j^{[r]}(x_j) &= \hat{m}_j(x_j) \ominus \bar{Y} \\ &\ominus \sum_{k < j} \int_{\mathcal{X}_k} \hat{f}_k^{[r]}(x_k) \frac{\hat{p}_{jk}(x_j, x_k)}{\hat{p}_j(x_j)} d\mu_k(x_k) \\ &\ominus \sum_{k > j} \int_{\mathcal{X}_k} \hat{f}_k^{[r-1]}(x_k) \frac{\hat{p}_{jk}(x_j, x_k)}{\hat{p}_j(x_j)} d\mu_k(x_k). \end{aligned}$$

4 The main results

The paper establishes the existence and uniqueness of the semimetric Bochner smooth backfitting estimator and proves convergence of the iterative algorithm. For Hilbertian and Riemannian predictors, it derives convergence rates for both

$$\hat{f} = \bar{Y} \oplus \sum_{j=1}^d \hat{f}_j$$

and the individual component estimators $\hat{f}_j$.

For Hölder-continuous component maps, the convergence rates depend on the individual predictor dimensions q_j , rather than on the total dimension

$$\sum_{j=1}^d q_j.$$

Thus, the convergence rate does not depend on the total predictor dimension in the same way as a full-dimensional nonparametric estimator. The paper also provides a more general convergence result for predictors in arbitrary semimetric spaces. Simulation studies and real data analyses illustrate the use of the method with mixed Euclidean and non-Euclidean predictors and Hilbert-space-valued responses.

5 My takeaways

- Which part of the smooth backfitting argument makes the convergence rate depend on each individual dimension q_j , rather than on the total dimension $\sum_{j=1}^d q_j$?

- How does strong dependence among the predictors affect the existence, uniqueness, and convergence of the backfitting solution?
- Which parts of the method would fail if the response took values in a general metric space rather than a Hilbert space?

References

- [1] J. M. Jeon, B. U. Park, and I. Van Keilegom. Additive regression for non-Euclidean responses and predictors. *The Annals of Statistics*, 49(5):2611–2641, 2021. <https://doi.org/10.1214/21-AOS2048>.